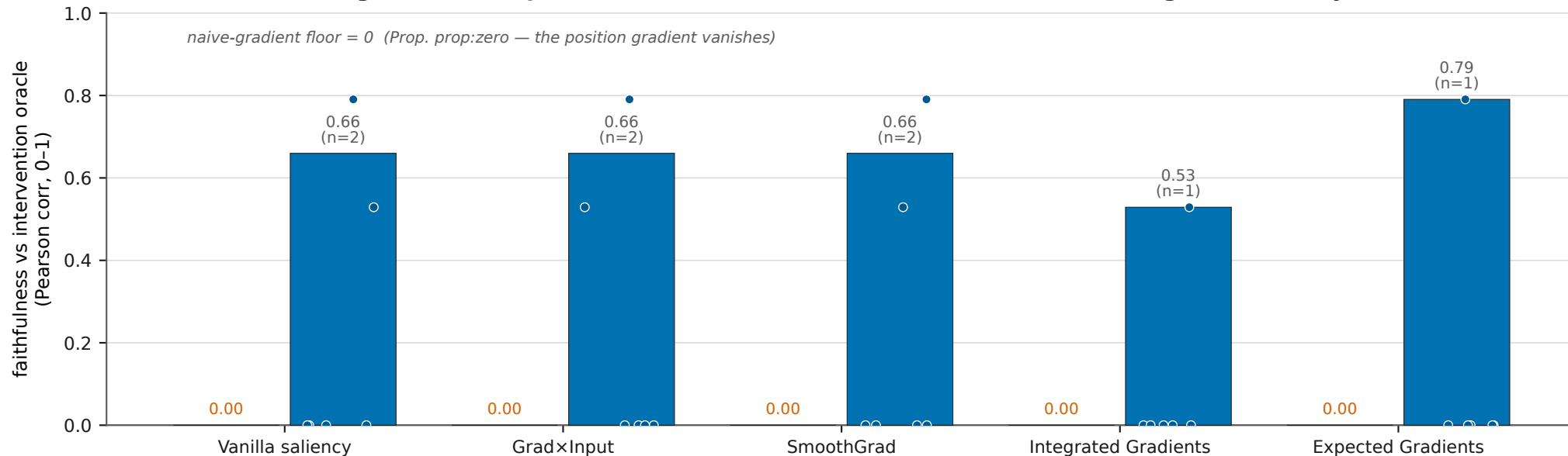


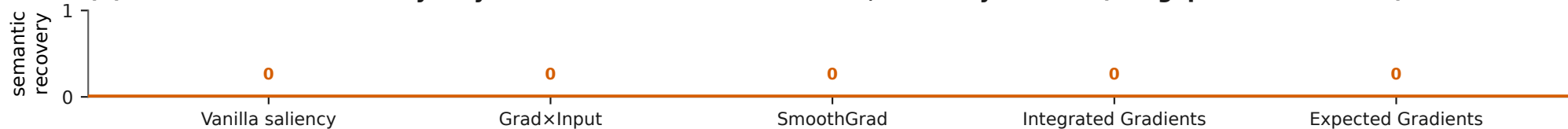
Keystone: the bilinear sampler makes gradient attribution faithful on the position regime — yet recovers no semantics

On the discrete position/index regime the naive gradient is provably zero (Prop. prop:zero), so every gradient method scores faithfulness 0. Turning Paper-1's bilinear index-boundary sampler ON (same surrogate as the SI joystick tool) restores a real position gradient and the methods become FAITHFUL — up to 0.791 (Vanilla saliency on pong). Yet semantic_recovery stays 0 for all 30 (method × game) records in BOTH conditions: an attribution map names no game concept on its own. Faithfulness is repairable; the semantic gap is not — the danger zone is not a fixable artifact of the vanishing gradient.

(a) Position/index regime: the sampler RESTORES faithfulness (0 → off the floor) for the gradient family



(b) ...but semantic recovery stays flat at 0 — in BOTH conditions, for every method (the gap survives the fix)



■ naive gradient (sampler off) — the \$1 floor ● per-game sampler faithfulness (sampler_faithfulness.csv; 6 core games, 120+30-frame state; pure read — no experiment re-run).
■ bilinear sampler ON — faithfulness restored — semantic recovery = 0 (flat, both conditions)